

STAR CLUSTERS

A LARGE GROUP OF STARS BOUND TOGETHER BY GRAVITY—ANYTHING FROM A DOZEN TO SEVERAL MILLION STARS—IS CALLED A STAR CLUSTER. THE MILKY WAY GALAXY CONTAINS THOUSANDS OF THESE SPECTACULAR STAR AGGREGATIONS.

Star clusters fall into two types: globular and open. Globular clusters are ancient, dense cities of stars, some containing more stars than a small galaxy. Open clusters, in contrast, are young, contain far fewer stars, and are often the site of new star creation. Many open clusters, and a few globular ones, can be seen in the night sky with the naked eye. Both types can be a magnificent sight when viewed through binoculars or a telescope.

Globular Clusters

Globular clusters are groups of between 10,000 and several million mostly very old stars arranged roughly in a sphere. More than 150 clusters like this exist in the Milky Way—each can last for 10 billion years. The stars in a cluster tend to be concentrated toward its center and move in random circular orbits around the center.

Many globular clusters consist of a single population of stars that all have the same origin, similar ages, and chemical composition. However, some contain two or more populations that formed at different times—through some of the more massive stars in the initial population dying and materials from them being recycled into a second star generation.

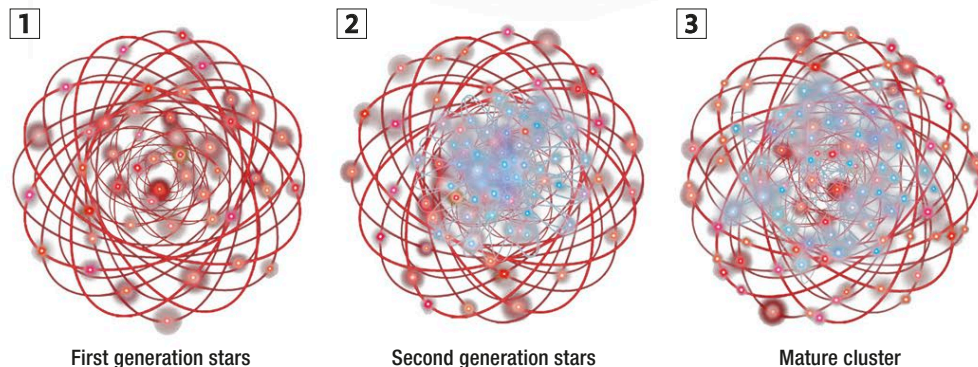
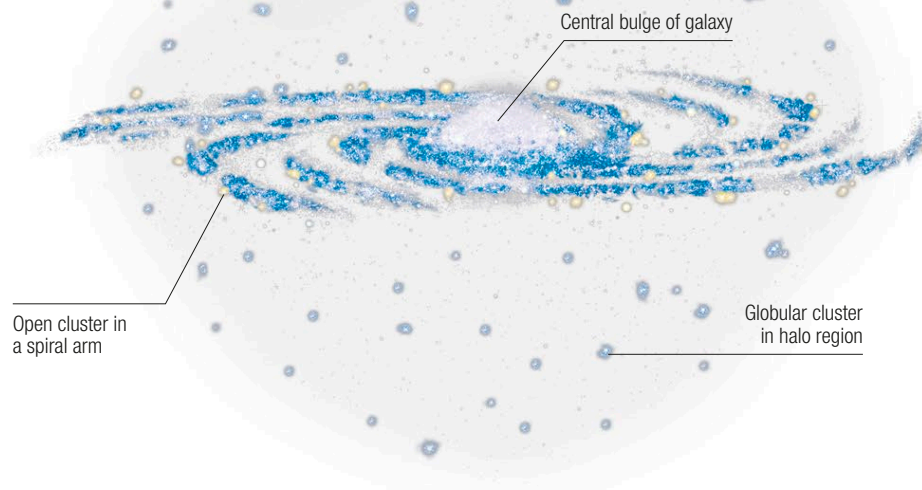
Open Clusters

Open clusters are groups of up to a few thousand stars that were formed roughly at the same time from the same cloud of gas and dust. They are more loosely bound by gravity than globular clusters, and they survive for a shorter time—from a few hundred million up to a few billion years. Unlike globular clusters, which occur in all types of galaxies, open clusters are found only in spiral and irregular ones, where stars are actively being created. Around 1,100 clusters of this type have been identified so far in the Milky Way.

Our galaxy's **largest globular cluster**, called **Omega Centauri**, contains about **10 million stars**

▽ Cluster distribution in spiral galaxies

Globular and open clusters exist in different parts of spiral galaxies like the Milky Way—globular clusters in the halo region, above and below the main disk, and open clusters in the galaxy's disk and spiral arms.



△ Evolution in a globular cluster

In this example of cluster evolution, some of the first star generation (red) die. Material from them then forms a second generation (blue), more concentrated at the centre of the cluster. Gradually their orbits change, mixing them with the older red stars.



△ M7 open cluster

Also known as the Ptolemy cluster, this array of around 80 stars lies in the constellation of Scorpius. Although 980 light-years away, it is easily seen with the naked eye.